

✦ Geospatial Analysis: Frequent Locations & Movement Patterns

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Project Motivation

- Geospatial data is time-based data that corresponds to a location on Earth determined by its coordinates.
- ✦ • This project aims to analyze a dataset containing location records for 20 individuals to recognize patterns and identify frequently visited locations.
- This analysis could be useful for estimating energy usage and traffic levels, based on how and when people tend to travel.

Project Overview

- The dataset contains time-stamped location coordinates, keeping a record of a person's location on a certain date and time.
- ✦ • From this, we can map out a person's daily schedule and movement patterns.
- We can also estimate their home and work locations, based on the amount of time spent and what time of day they visit that location.
- Lastly, this analysis can then be used to train a predictive model that can guess a person's location based on the time and day of the week.

Implementation I

- Goal: Find the top 5 locations in a month and estimate home and work location for each individual.
- ✦ • Use DBSCAN clustering to find clusters where most time is spent.
- Begin with:
 - Tidy dataset (handle missing values and drop rows with low accuracy)
 - Explore dataset (find number of locations visited in a month and most visited locations)

Implementation II

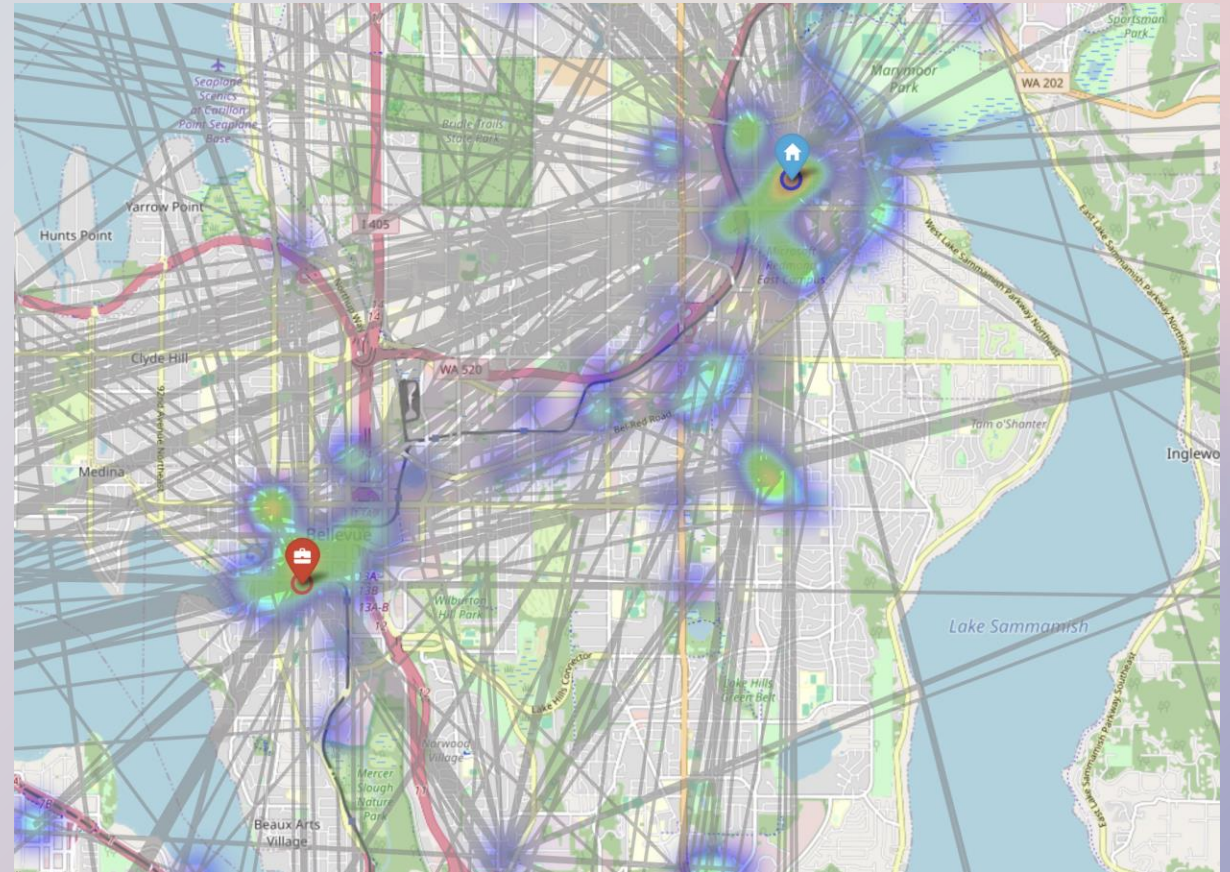
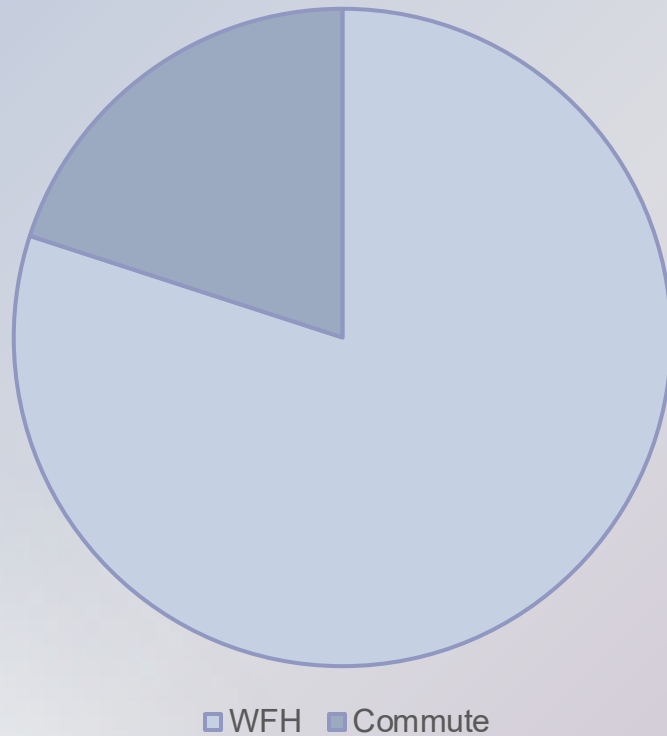
- Now we start clustering,
 - ✦ ○ Downsize dataset (Sort into 30min interval time buckets)
 - Project coordinates to meters
 - DBSCAN using Euclidean distance
 - Find Top 5 locations per month (based on time spent in each cluster)

Implementation III

- We can now use the clusters to estimate home and work locations.
- To do this,
 - ✦
 - Categorize times as “night”, “workday” or “other”
 - Most time spent in a cluster at night = home location
 - Most time spent in a cluster during the workday = work location
- Lastly, create a heatmap to visualize the clusters

Analysis Results

Workplace of Individuals
Analyzed



Map for Client 336 living Redmond/Bellevue area

Model Implementation

- Train a general predictive model for individuals living in Portland/Vancouver area who work from home.
- ✦ • Given a time, give the category of the cluster they are most likely to be at.
- Uses a Naïve Bayes Classifier – predict the probability of the cluster that the individual is more likely to be in
- Demo output:

```
Datetime: 2024-11-18 04:30:00  
Predicted category: home  
Datetime: 2024-11-18 10:30:00  
Predicted category: work  
Datetime: 2024-11-18 05:30:00  
Predicted category: home
```

Thank You!